

Surgery & Allied Subjects

Departmental Objectives

The aim of this course is to provide community oriented & need based education so as to produce basic doctors who will be able to:

- elicit a complete clinical history & physical findings and formulate diagnosis of common surgical problems prevalent in Bangladesh and abroad.
- carry out necessary investigations & interpret the results with proper utilization for management
- perform minor surgical procedures and treat minor surgical problems
- recognize the major surgical problems needing specialized care, initiate the primary treatment and refer to the appropriate centers
- diagnose and provide competent primary care in surgical emergencies.
- carry out the responsibility of management in common casualties or natural calamities to offer and arrange basic life support.
- take necessary preventive & prophylactic measures for surgical problems
- be involved in continued care & rehabilitation of surgical patients.
- deliver health education in the community with emphasis to the preventive aspects of surgical disorders.
- demonstrate the right attitude in
 - ❑ Patient Care
 - ❑ Community health care
 - ❑ Continuing medical education & research
 - ❑ Observing the moral & legal codes of medical ethics

List of Competencies to acquire:

1. Clinical –

- a. rapport building with patients, relatives, colleagues, health care professionals and supporting staffs of the hospital
- b. take detail relevant history
- c. conduct thorough clinical Examination
- d. decide on a provisional working diagnosis
- e. perform and/or order relevant investigations considering the cost effectiveness
- f. interpret common laboratory and imaging investigations
- g. calculate fluid and electrolyte requirements
- h. evaluate and make initial management of acute trauma patient
- i. adopt aseptic techniques and procedures and maintain principles of sterilization

2. Communication-

- a. obtain permission before any examination and clinical procedures
- b. obtain informed consent for surgical procedures including organ ablation.
- c. appreciate right to privacy and information about the disease and its consequence

3. Managerial-

- a. provide leadership during team work
- b. implement time management skills
- c. issue certificates (discharge, death, medical and injury).
- d. write notes (case notes, operation notes, referrals)
- e. keep detail and systematic records both manual and electronic
- f. use computer and IT facilities.

4. Manipulative and practical skills-

- a. adopt universal aseptic techniques in handling surgical patient
- b. start IV lines
- c. insert NG tubes
- d. introduce urethral catheter and perform supra-pubic cystostomy
- e. drain superficial abscesses
- f. perform per-rectal examination
- g. achieve emergency control of revealed hemorrhage
- h. carry out initial management of wound
- i. repair minor wounds
- j. complete primary management of fractures and arrange transfer to appropriate centers.
- k. apply splints, slings, POP casts and slabs, tractions, bandages, sterile dressings

Distribution of teaching - learning hours Surgery & Allied Subjects

Subject	Lecture (in hours)				Small group teaching (in hours)	Departmental integrated teaching (in hours)	Phase integrated teaching (in hours)	Clinical/Bedsid e teaching (in weeks)			Total weeks	Block posting (in weeks)	Formative examination (in days)		Summative examination (in days)	
	2 nd Phase	3 rd Phase	4 th Phase	Total	PBL, Practical demonstration , Instrumental demonstration, Skill lab, Tutorial & etc.			2 nd Phase	3 rd Phase	4 th Phase			Preparatory leave-10 days	Exam time-15days	Preparatory leave-10 days	Exam time-30days
General surgery	35	30	60	125	134 hours	(11 topics ×2 hours) = 22 hours	(42 topics × 3 hours) = 126 hours	15	01	07	23	04 wks	Preparatory leave-10 days	Exam time-15days	Preparatory leave-10 days	Exam time-30days
Orthopaedic surgery	-	15	45	60				02	04	04	10					
Radiology	-	-	05	05				01	-	-	01					
Radiotherapy	-	-	08	08				-	01	-	01					
Anesthesia	-	10	-	10				01	-	-	01					
Neurosurgery	-	-	05	05				-	01	-	01					
Pediatric surgery	-	05	10	15				-	-	02	02					
Urology	-	05	10	15				-	-	02	02					
Burn & Plastic surgery/ Emergency & Casualty	-	-	05	05				-	-	01	01					
Dentistry	-	-	-	-				01			01					
Ophthalmology	-		38	38				-	04	04	08					
Otolaryngology	-		38	38				-	04	04	08					
Total	324				134	22	126 hrs	20	15	24	59 wks	04 wks	25 days		40 days	
Grand Total	480 hours						126 hrs	63 weeks					65 days			
Time for integrated teaching, examination preparatory leave and formative & summative assessment is common for all subjects of the phase																
Preventive aspects of all diseases will be given due importance in teaching learning considering public health context of the country and others parts of the world.																
Related behavioral, professional & ethical issues will be discussed in all clinical and other teaching learning sessions																

Surgery & Allied Subjects: Hours distribution for Clinical/Bedside teaching in 2nd, 3rd & 4th phases in details

Subject	Clinical/Bedside & Ambulatory care teaching (in hours)						Total hours (in three phases)	Total weeks {(2 nd phase wks + 3 rd phase wks + 4 th phase wks = Total three phases wks) × (6 days × 4 or 2 hours)}
	2 nd Phase		3 rd Phase		4 th Phase			
	Indoor clinical/ bedside teaching & Ambulatory care teaching		Indoor clinical/ bedside teaching & Ambulatory care teaching		Indoor clinical/ bedside teaching & Ambulatory care teaching			
	Morning	Evening	Morning	Evening	Morning	Evening		
	Indoor/ OPD/ Emergency & Casualty	Indoor/ Emergency & Casualty	Indoor/ OPD/ Emergency & Casualty	Indoor/ Emergency & Casualty	Indoor/ OPD/ Emergency & Casualty	Indoor/ Emergency & Casualty		
	21 weeks		15 weeks		24 weeks			
General surgery	180 h (15w)	180 h (15w)	12 h (1w)	12 h (1w)	84 h (7w)	84 h (7w)	552 h	(15+01+07) = 23 w × (6 days × 4 hrs)
Orthopaedic surgery	24 h (2w)	24 h (2w)	48 h (4w)	48 h (4w)	48 h (4w)	48 h (4w)	240 h	(2+4+4) = 10 w × (6 days × 4 hrs)
Radiology	12 h (1w)	-	-	-	-	-	12 h	(1+0+0) = 01 w × (6 days × 2 hrs)
Radiotherapy	-	-	12 h (1w)	-	-	-	12 h	(0+1+0) = 01 w × (6 days × 2 hrs)
Anesthesia	12 h (1w)	12 h (1w)	-	-	-	-	24 h	(1+0+0) = 01 w × (6 days × 4 hrs)
Neurosurgery	-	-	12 h (1w)	12 h (1w)	-	-	24 h	(0+1+0) = 01 w × (6 days × 4 hrs)
Pediatric surgery	-	-	-	-	24 h (2w)	24 h (2w)	48 h	(0+0+2) = 02 w × (6 days × 4 hrs)
Urology	-	-	-	-	24 h (2w)	24 h (2w)	48 h	(0+0+2) = 02 w × (6 days × 4 hrs)
Burn & Plastic surgery/ Emergency & Casualty	-	-	-	-	12 h (1w)	12 h (1w)	24 h	(0+0+1) = 01 w × (6 days × 4 hrs)
Dentistry	12 h (1w)	-	-	-	-	-	12 h	(1+0+0) = 01 w × (6 days × 2 hrs)
Ophthalmology	-	-	48 h (4w)	48 h (4w)	48 h (4w)	48 h (4w)	192 h	(0+4+4) = 08 w × (6 days × 4 hrs)
Otolaryngology	-	-	48 h (4w)	48 h (4w)	48 h (4w)	48 h (4w)	192 h	(0+4+4) = 08 w × (6 days × 4 hrs)
Total	240 hrs	216 hrs	180 hrs	168 hrs	288 hrs	288 hrs	1380 hrs	59 weeks

Teaching-learning methods, teaching aids and evaluation

Teaching Methods				Teaching aids	In course evaluation/ Formative
Large group	Small group teaching	Self learning	Others		
Lectures	Tutorials, Problem Based Learning, Clinical demonstrations OPD / indoor attending & observing minor operations Demonstrations of X-rays specimen, Observations in ICU, Postoperative ward, Case Presentation and discussion. Skill lab practice	Assignment, Self study	Integrated teaching, Visit to radiotherapy Attend centers where investigations for hearing impairment, vertigo, Tinnitus are available.	Computer, Chalk & board, OHP, Multimedia, Photographs & Videos, Specimens, & Models, Plain & Contrast X-rays of Upper & lower GIT, I.V.U, Fractures Skull X-rays Sinogram & Fistulogram Ultrasonography, Abdomen HBS & Pancreas Urinary tract Scans, thyroid scans, C.T. Scan, MRI, PET Scan, Bone scan, Doppler and duplex imaging. Immunohistochemistry	Item Examination Card final, Term Examination Term final (written, oral+ practical + clinical) Marks distribution: a) Surgery— I. Card final-3 II. Term Final-2 III. MCQ of Integrated teaching-5 b) Ophthal- 5 c) ENT-5

Final professional Examinations:

Marks distribution for assessment of surgery

Total marks Surgery and allied Subjects – 500

- Written = 200 (Formative Assessment-20 +(MCQ- SBA & MTF) 40+ (SAQ +SEQ) 140 =200)
- Structured Oral = 100 (60+20+20)
- Clinical = 100 (60+20+20)
- Practical (OSPE/OSCE) = 100 (60+20+20)

Total in Surgery and allied---500.

Related Equipments:

General surgery	Materials
a. Sets -butterfly needle & cannula, Infusion and Transfusion sets b. Tubes--Feeding tubes, NG tube, Flatus tube , ‘T’ tube, Chest drain set, Endo-tracheal tube c. Bags- Blood bags, Stoma bags, Fluid bags, Nutrition bags, Urine bags, Drain bags, Bi-channel d. Sharps- BP blade and handle, surgical scissors, Needle holder, Surgical suture materials, Gloves, gown, mask, caps, surgical goggles e. Forceps-Sponge holding forceps , towel clip, Alli’s tissue forceps, artery forceps, Sinus forceps, dissecting forceps, Kocher’s artery forceps, kidney tray, gully pot, intestinal clamps, f. Retractors—Deavers, abdominal, Morris abdominal retractor, Langhanbach’s retractor,	g. Special-Lane’s twin gastro jejunostomy clamp, proctoscope, metallic urethral dilators, nephrolithotomy forceps, Bone nibbler, Osteotome, chisel, hammer, amputation saw, SPC set, CV line set, Spinal needle, h. Orthopedic--Plaster of Paris bandage, crepe bandage, Splints supporting aids- Cervical collar, Circle brace, artificial limb, i. Anesthesia- machine, Laryngoscope, airway tube, Umbo bag, pulse oximetry, Digital Thermometer, Oxygen cylinder with devices (These equipment may be used in OSPE procedure stations)
ENTD	
Thudicum nasal speculum, Killians self retaining nasal speculum, Lichwitz antrum puncture trocar and cannula, Higginson’s rubber syringe, Walsham’s forceps, Luc’s forceps, Tilley’s forceps, St Clair Thomson post nasal mirror, Jobson horn probe and ring curette, Tuning fork, Head mirror,	Boyle Davis mouth gag, Luc’s tongue depressor, Draffins bipod metallic stand, Eve’s tonsillar snare, St Clare Thomson Adenoid curette and cage, Trousseau’s tracheal dilator, Jackson’s metallic tracheostomy tube, Direct laryngoscope Chevalier Jackson’s oesophagoscope, Negus bronchoscope etc.
Ophthalmology	
Trial lens, trial frame, Eye speculums (Wire, Universal), DCR punch, Tonometer, Ophthalmoscope, Cat’s paw retractor, BP Blade & handle, Keratome, Squint hook	Iris repositor, lens dialer, two way cannula, chalazion clamp and scoop, corneal forceps, irrigating vectis, sac guard, sac dissector, lacrimal probe, punctum dialtor etc.

Learning Objectives and Course Contents in Surgery

Learning Objectives	Contents	Teaching Hours
A. Basic and Principles of Surgery Student should be able to: - state the history , evolution and scope of Surgery - assess and prepare patient for surgery - understand the patho-physiology of trauma - diagnose, treat and manage minor wounds - diagnose, treat and manage surgical infections (boil, abscess, carbuncle & gangrene) . - diagnose and provide basic treatment for shock & haemorrhage. - recognize all external hernias & their complications & initiate primary care for complicated hernias. - recognize & differentiate different types of burns and initiate primary care & take measure to prevent complications. - recognize fluid & electrolytes imbalance states, investigate & initiate appropriate therapy. - recognize, & investigate different types of skin ulcerations. - recognize, investigate & treat superficial skin tumour & cysts - take appropriate measures to prevent hospital infection. - understand and comply with ethical principles in clinical practice	CORE Phase II - History, evolution and scope of surgery - Approach to a surgical patients - Surgical diagnostic process and techniques - Surgical Infection (Boil , Furuncle, Abscess, Carbuncle , cellulites) - Septicemia (causes, complications and treatment) - Sinus, Fistula and cysts - Wounds (classification and management) - Ulcers , pressure sores - Groin hernias - Haemorrhage - Shock	20 hours
	Phase III - Metabolic response to injury - Principles of Management of Trauma - Management of a severely injured patient - Fluid and electrolytes balance - Enteral and Parenteral nutrition	10 hours
	Phase IV - Pre operative assessment and preparation - Tumours of skin - Lymphadenopathy (causes, investigations, diagnosis, biopsy) - Surgical ethics	10 hours
	ADDITIONAL Organ transplantation, Robotics in surgery	

Learning Objectives	Contents	Teaching Hours
B. Systemic Surgery 1. Alimentary System Student should be able to : - investigate and diagnose the common surgical diseases of alimentary system and suggest management - diagnose the acute conditions of alimentary system and initiate primary care - identify the patient requiring specialty surgical intervention & refer to appropriate centre - take continued care of the operated patients - recognise post operative complications & take appropriate measures.	CORE Phase II Complications of Peptic ulcer (Perforation, Pyloric stenosis) Upper G.I. Tract bleeding Appendicitis Intestinal obstruction; Phase III Abdominal trauma (Diagnostic and Management principles) Ruptured Spleen Ruptured liver Ruptured intestine Phase IV Tongue, Lip & other oral lesions (ulcer, cancer) Oesophagus Carcinoma oesophagus and stricture Carcinoma stomach Neoplasm of colon and rectum Intestinal tuberculosis Anal canal Haemorrhoids, Fistula, Sinus & Fissure, Carcinoma anus Colostomy & ileostomy (indications and management) Abdominal incisions (Tutorial) ADDITIONAL Intra abdominal abscess Diseases of salivary glands Hiatus hernia.	5 hours 5 hours 5 hours

Learning Objectives	Contents	Teaching Hours
2. Genito-Urinary System Student should be able to- 1. diagnose common congenital G.U. anomalies & advise / refer to appropriate centers 2. diagnose and manage acute GU conditions like • Acute retention of urine • Acute epididymo- orchitis • Torsion testis • Paraphimosis • Phimosis • Acute ureteric colic • Urosepsis 3. evaluation of scrotal swelling 4. evaluate a case of haematuria 5. order necessary investigations, and interpret the result of investigation & suggest principles of management 6. recognize a case of retention of urine , find out causes perform aseptic catheterization 7. introduce suprapubic catheter 8. describe the steps of circumcision	CORE Phase III 1. Urinary symptoms & definitions 2. Urological investigations and their interpretations, 2. Developmental genito-urinary anomalies 3. Scrotal swelling • Hydrocele • Scrotal cellulitis 4. Acute scrotal conditions • Epididymo- orchitis • Torsion testis Phase IV 5 Urolithiasis (Causes ,Diagnosis , Principles and modalities of treatment) 6 Retention of urine (acute and chronic 7 Hydronephrosis 8 UTI 9 Urinary tract injury. • Renal injury • Urethral injury 10. Renal Neoplasm • RCC • Wilm's Tumour 11 Testicular Tumour 12 BPH 13 Stricture urethra <u>ADDITIONAL</u> • Male infertility • Minimal Invasive Surgery in Urology	20 hours 10 hours

Learning Objectives	Contents	Teaching Hours
3 Hepatobiliary & Pancreas Student will be able to: 1. diagnose, investigate cholecystitis, cholelithiasis & Choledocholithiasis 2. suspect pancreatitis; initiate primary case management & suggest management 3. investigate & interpret the results in case of obstructive jaundice & suggest appropriate treatment 4. diagnose & investigate suspected case of liver & sub-phrenic abscess & suggest appropriate treatment.	CORE Phase II Cholelithiasis (causes and complications) Cholecystitis (acute & chronic) Pancreatitis (acute pancreatitis) Phase IV Obstructive jaundice Pancreatic tumours Liver abscess ADDITIONAL Hepatic neoplasm Cysts of liver Neoplasm of Gall Bladder	5 hours 5 hours 4 hours
4 Endocrine & Breast Students will be able to: 1. assess, investigate & diagnose thyroid swelling & thyrotoxicosis and suggest principles of management 2. diagnose & manage a case of breast abscess 3. assess, investigate & interpret the status and diagnose a case of breast lump & suggest principles of treatment.	CORE Phase IV Thyroid Goitre and Neoplasms of thyroid Breast Breast pain, Mastitis and Breast Abscess Fibro-adenosis and Fibroadenoma Carcinoma of breast ADDITIONAL Diseases of adrenal gland Diseases of Parathyroid gland	4 hours 4 hours 2 hours

Learning Objectives	Contents	Teaching Hours
5 Chest Students will be able to: • assess & diagnose traumatic haemopneumo-thorax, associated injuries & introduce water seal drain in appropriate case.	<u>CORE</u> <i>Phase IV</i> Chest injury (Haemothorax, Pneumothorax) Chest tumours, Chest drain, <u>ADDITIONAL</u> Dysphagia Empyema thoracis	3 hours
6. Cardio-vascular System Students will be able to: 1. recognize chronic ischaemic conditions of limbs 2. take appropriate preventive measures & refer to specialized centre. 3. take appropriate measure to prevent DVT 4. recognize early cases of DVT 7. Plastic & Reconstructive Students will be able to 1. manage Burn patient and minimize their complications 2. take any major wound care 3. suggest measures for con. External deformity & disfiguration	<u>CORE</u> <i>Phase III</i> Vaso occlusive disorders Atherosclerosis, Buerger's disease Varicose vein Thrombophlebitis Deep vein thrombosis <u>ADDITIONAL</u> Pulmonary embolism Angioplasty, CABG and cardiac surgery <u>Core</u> <i>Phase IV</i> Burn (Causes, complications and management) Skin grafting Skin tumours, Special area burn , Inhalation and electric burn	5 hours 5 hours

Learning Objectives	Contents	Teaching Hours
8. Neuro surgery Students will be able to: 1. provide primary care of head injury & Spinal injury cases. 2. take measures to prevent complications in neuro surgical patients. 3. involve effectively in continued care & rehabilitation of neuro surgical cases.	<u>CORE</u> Phase IV Head injury ICSOL PLID Paraplegia/hemiplegia <u>ADDITIONAL</u> Hydro cephalus Tumours of brain Tumours of spinal cord	5 hours
9. Operative Surgery Student should be able to perform: 1. primary & delayed primary & Secondary suture closure of wounds 2. Circumcision 3. Vasectomy 4. drainage of superficial Abscess 5. Venesection 6. Hydrocele operation 7. excision of superficial cysts & tumours 8. dressing of surgical wounds	<u>CORE</u> Phase III Principles of Asepsis & Antisepsis Pre-operative assessment & preparation Venus access Circumcision Operation for hydrocele Repair of D.U perforation Wound care <u>Tutorials</u> Universal precautions (Scrubbing , gloving & gowning) O.T. environment & behavior Preoperative skin preparation and draping Suturing materials ,Stitches	5 hours 5 hours

Learning Objectives	Contents	Teaching hours
Student should be able to : assist in common major operations & take post operative care	Phase IV Common Abdominal incision Operation for inguinal hernia Drainage of abscesses Catheterisation , Supra-pubic cystostomy Anastomosis Appendicectomy Cholecystectomy Gastrojejunostomy Basic principles of Laparoscopy. Additional Thyroidectomy, Nephrectomy, Mastectomy / Prostatectomy	10 hours
10. Orthopedic Surgery Student should be able to: - apply ATLS protocol to provide resuscitation of polytrauma patient. - manage simple and undisplaced fractures - demonstrate skill in wound excision of open fractures . - demonstrate skill in: - application of splints, slings , traction. - application of plaster cast and slab - plaster techniques and design - versatility & possible complications of plaster - the art of application of plaster & its' removal - manipulative reduction of common fracture and dislocation. - aseptic technique of joint fluid aspiration. - diagnose and outline treatment for acute osteomyelitis and septic arthritis - identify patient for referral to appropriate centre - demonstrate knowledge and understanding of the basic principle of physiotherapy and rehabilitation.	CORE Phase III a) General Orthopedics - Introduction to orthopaedics - Hard tissue trauma :- - Fracture classification - Principal of management of open and closed fracture - Fracture healing –nonunion, malunion, delayed union. - Infection of bone (Acute and chronic osteomyelitis) Phase III b) Regional orthopedics Upper limb Colles' fracture Supracondylar fracture Clavicle fracture Radius Ulna fracture (Shaft) Humerus fracture (Shaft) Lower limb Fracture of Shaft of femur Fracture of Tibia fibula	5 hours 10 hours

Learning Objectives	Contents	Teaching Hours
	Phase IV Regional Orthopaedics • Upper Limb Hand injuries and Hand Infection • Lower Limb Fracture of Neck of femur Fracture of Pelvis Ankle and foot injuries Amputations Additional Dislocation – Hip, Haemarthrosis • Soft tissue trauma (muscle and tendon injuries, compartmental syndrome) • Infection of joint including osteoarticular tuberculosis • Mass Casualty- ATLS, Disaster management. • Bone tuberculosis Additional Dislocation of shoulder and elbow b) Paediatric orthopedics : Congenital anomalies-talipes, DDH, Bow legs, Polydactyly, Claw c) Bone tumors : Classification of bone tumor Common benign and malignant bone tumor – osteochondroma, Giant cell tumor, Osteosarcoma, Metastatic bone tumor. Vertebral fracture – (primary management, transportation. Principles of definitive management) Additional d) Tendinitis, Tenosynovitis, bursitis.	45 hrs

Learning Objectives	Contents	Teaching Hours
11. Anaesthesiology Student should be able to : • be aware of the safety in Anaesthesia. • be aware of the possible complications & management • demonstrate basic knowledge and perform Cardio-Pulmonary Resuscitation (CPR) • describe the scope of Anaesthesia in rural environment. Practical Skills Student should be able to perform : • pre-operative assessment • induction • intubation • I/V line • artificial ventilation • post-operative room care	Phase III CORE a) Anesthesia as a subject: its scope, outline- present & future b) Anesthesia Pharmacology: Drugs: induction, maintenance, muscle relaxants c) Intra-operative management d) Post-operative management and complication e) General GAnes (G.A) f) Local/Regional anesthesia g) Management of Pain (chronic) h) Intensive Care Unit (ICU) i) Basic life support. j) Cardio-Pulmonary Resuscitation (CPR) Exposure to practical procedures (Tutorial) : • Pre-operative assessment • Induction • Endo tracheal Intubation • CV line • Artificial ventilation • Face mask ventilation. • Recovery room experience	10 hours

Learning Objectives	Contents	Teaching Hours
12. Radio Diagnosis & Imaging Student should be able to : • demonstrate knowledge and understanding of the principles of radiology and imaging • appreciate the importance of imaging as investigation & diagnosis of clinical conditions • describe the hazards of radiation • describe the protection measures for personal patient and the community. • write proper requisition for various x-rays & imaging. X-RAY Chest Student should be able to : • differentiate normal anatomical images from those due to pathological states, • diagnose the common conditions like tuberculous consolidation, pleural effusion, pneumothorax, lung abscess, collapse, bronchogenic carcinoma. • make radiological diagnosis of mediastinal masses	<u>CORE</u> Phase IV • Introduction of radiology & imaging including CT & MRI • Hazards of radiation and protection for personals, and patients. • Principles of ultra-sonography & its clinical application • Plain & contrast X-Rays • Interventional imaging • USG <u>CORE:</u> • Normal and pathological image • Pneumonic and Tuberculous consolidation • Pleural effusion • Pneumo Thorax <u>Additional</u> • Lung abscess • Mediastinal mass	6 hours 2 hours

Learning Objectives	Contents	Teaching Hours
<i>Gastro intestinal system</i> Student should be able to : - diagnose intestinal obstruction, perforation etc. - recognise indications and contra-indication for barium studies e.g. meal, swallow, follow-through & enema. - make differential diagnosis of stones & calcification on plain X-Ray. - diagnose gastric ulcer, duodenal ulcer, growth in the stomach, oesophageal cancer on barium studies. - interpret the finding of cholangiogram.	<u>Core:</u> - Plain X-ray findings of Acute abdomen. - Indications & contraindications for barium studies. Hepatobiliary system Cholangiogram & ERCP - USG of HBS and Pancreas Additional: MRCP	
<i>Skeletal system</i> Student should be able to : - diagnose common fractures, dislocations & bone tumours bone infections with the help of X-rays <i>Excretory System</i> Should be able to : - identify renal calculi in plain X-ray - understand USG & IVU findings in renal stone and other renal diseases.	<u>CORE</u> - Diagnosis of common fractures of upper and lower limb - skull fractures - Spinal fractures and caries spine - Acute osteomyelitis - common bone tumours - diseases of joints - dislocations <u>CORE</u> - X-ray KUB & IVU - USG of Kidney, Ureter , Bladder and prostate	

Learning Objectives	Contents	Teaching Hours
13. Radiotherapy Students will be able to: • appreciate the role of radiotherapy in the management of cancer • demonstrate knowledge of radiation • identify different sources of radiation • refer the patients to radiotherapy department • recognize common radiation hazards after primary care Students will be able to: • recognise common cytotoxic drugs. • refer appropriate cases for chemotherapy. • recognise common complication & offer primary care.	Phase IV <u>CORE</u> <i>Introduction to Radiotherapy</i> Radiation oncology, basic principles and practices : • Aims of radiation oncology • Sources of radiation , Isotopes and their mechanism of action • Curative/Palliative radiotherapy • Radiosensitivity, radioresistance, radiocurability and normal tissue tolerance. • Common radiation reactions and management. Medical oncology, basic principles and practice : • Cell cycle and Mechanism of action of cytotoxic drugs • Clinical aspect of cancer chemotherapy • Complications of chemotherapy (Infection and bleeding tendency) • Chemotherapy of common cancers, • Common Chemotherapeutic regimes	5 hours

Learning Objectives	Contents	Teaching Hours
Students will be able to: • appreciate the role of doctors in prevention and early diagnosis of cancer & referral of cancer patients. • take leadership in the community to offer rehabilitative support • offer follow up & terminal care of cancer patients. • recognise clinical condition as which could be diagnosed by radio-isotope & interpret the results. • recognise diseases requiring isotope therapy.	Prevention of common cancer : • Primary prevention, Secondary prevention • Early diagnosis • Referral to appropriate centre Palliative support and terminal care : • Follow-up of cancer patients and terminal care Nuclear Medicine, basic Principles and practice : • Radio-isotope in diagnosis • Radio-isotope in therapy	1 hour 1 hour 1 hour

